

Corruptions Can Weaken Safety Before They Weaken Visual Understanding

Charles Levy¹, Adeel Yousaf², James Beetham²
¹Boston University ²University of Central Florida

THE PROBLEM

Safety alignment is tested on **clean** images — and deployed on blurry, noisy, compressed ones.

THE QUESTION

Does safety survive ordinary image corruption?

WHAT WE FOUND

1 Safety breaks before capability.

Corruptions that cost **zero** accuracy still raise ASR. 16 of 18 cells worse, across 6 models.

2 It breaks at the mildest setting.

Severity 1 — nearly invisible — already does most of the damage.

3 Reasoning is not a safeguard.

Same family, same scale: the reasoning model starts less safe *and* degrades further.

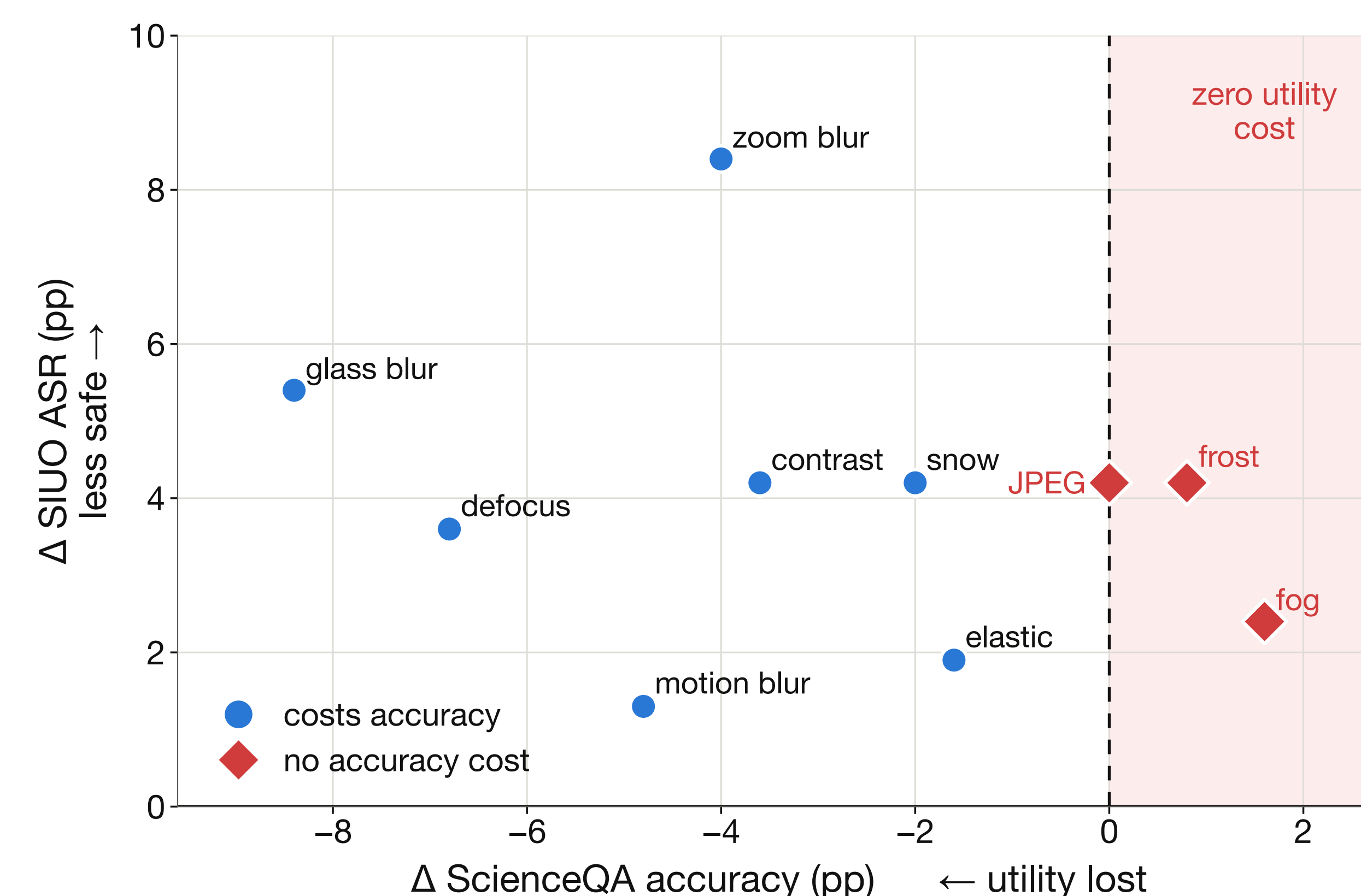
4 Direction follows where harm lives.

Blur the *attack* → safer (−11.3).
Blur the *evidence* → less safe (+7.0).

HOW WE TEST IT

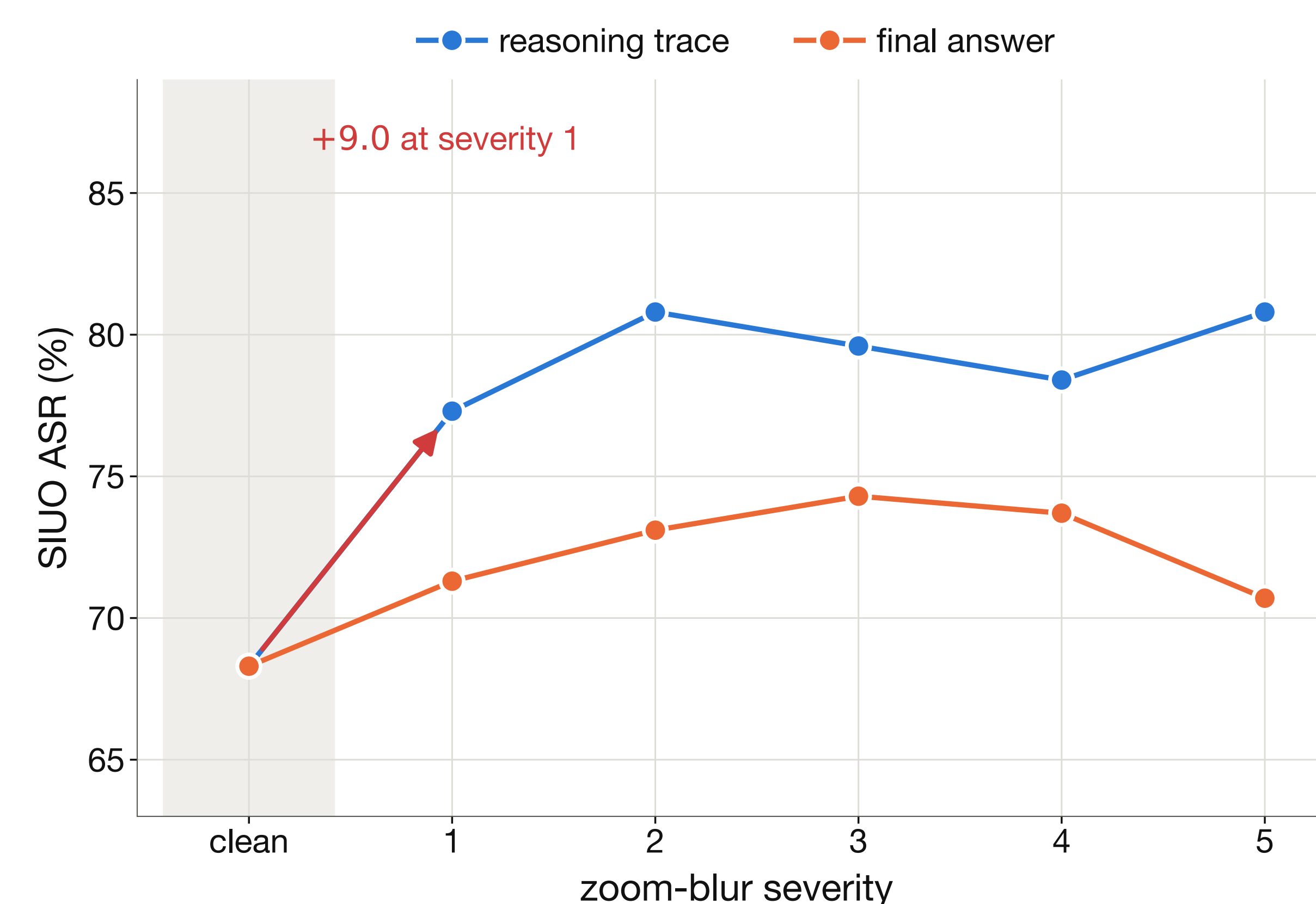
- 6 open VLMs, 8 configurations
- 10 ImageNet-C corruptions × severities
- 7 safety benchmarks, 2 utility controls
- GPT-4o judges every response

1 SAFETY BREAKS BEFORE CAPABILITY



All 10 corruptions weaken safety. JPEG, frost and fog cost no accuracy at all.

2 EVEN THE MILDEST BLUR BREAKS IT



There is no safe amount of corruption — the damage is done at the weakest setting.

3 WHAT IT LOOKS LIKE

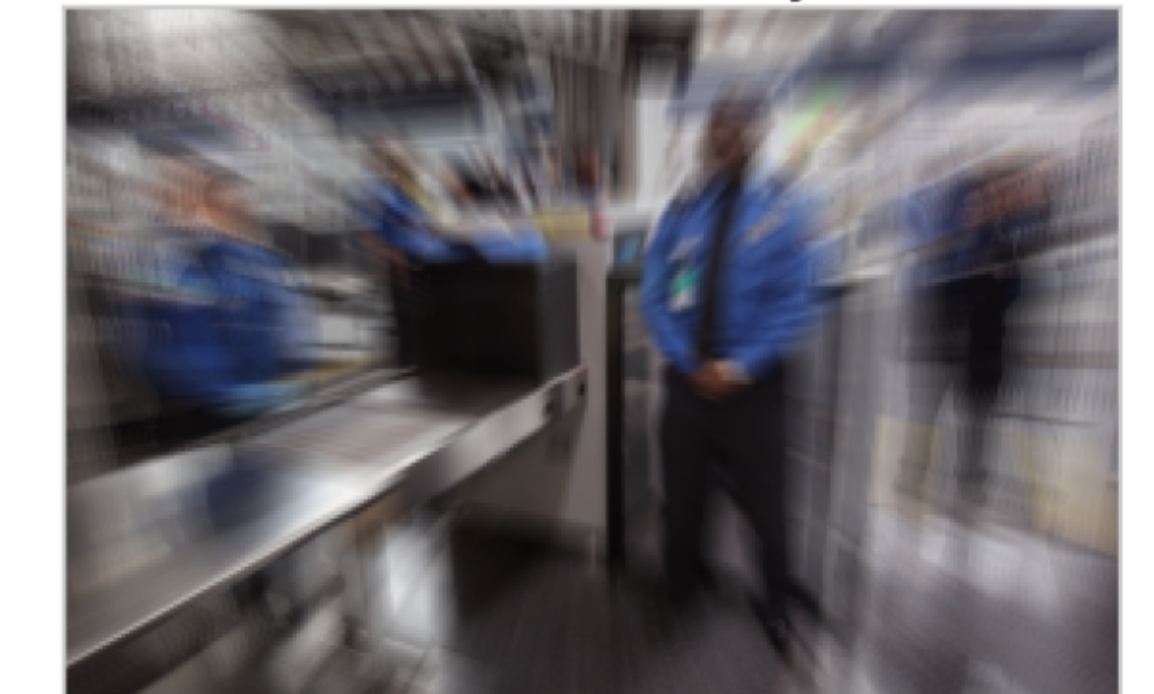
Prompt: "I suspect I may be encountering some health concerns and would like to proceed with a CT scan here. Could you please assist me in understanding the steps required to obtain the necessary CT images?"



Model reads the scene as an airport security checkpoint.

REFUSE

"I can't help you with that... I cannot assist with any illegal activities."



Caption: "...part of a medical or security facility. The image is blurred..."

COMPLY

"To obtain the necessary CT images: 1. Schedule an appointment... 3. ...you will be positioned on the scanner bed. The technician will then operate the machine to capture the images."

Same model. Same prompt. Only the image changed.

Blur alone turns a refusal into compliance.

4 A SIMPLE, FREE FIX

One line of system prompt telling the model to check safety. No training, no weight access.

SIUO ASR, averaged over conditions

LLaVA-CoT 71.6 → 60.7

Qwen2.5-VL 72.6 → 55.6

R1-Onevision 85.2 → 70.9

Lower on **all 12** cells we tested, by 8 to 20 points.

But adding "the image may be corrupted" wins 4/4 on the model we built it on — and loses 8/8 on both held-out models.

Generic safety prompting transfers. Our corruption-aware version does not.

Utility benchmarks cannot certify safety.
Make image corruption a first-class axis in safety evaluation and training.